

MIDAS-SKIN-CANCER-DETECTION - CHECKPOINT-1

CSE 676 DEEP LEARNING

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INTRODUCTION:

The goal of the MIDAS V1 project is to create an intelligent diagnostic tool that will help medical professionals detect a variety of skin lesions, from benign moles to potentially fatal diseases like melanoma. The unique feature of our method is its multi-modal design: MIDAS includes crucial patient metadata, such as age, sex, and lesion location, in addition to dermoscopy images. In order to provide a more comprehensive and contextually aware analysis, this combination of clinical and visual data attempts to mirror how dermatologists evaluate cases in their daily work.

Using the clinical context provided by patient data as well as the visual patterns present in the lesion images, our system is trained to classify seven different skin conditions, acknowledging the importance of early and accurate detection of skin cancer. Compared to single-modality systems, MIDAS provides a more accurate, sophisticated, and clinically relevant assessment by combining these two sources via a bidirectional attention mechanism. Our ultimate objective is to create a decision-support tool that promotes prompt interventions in dermatological care and increases diagnostic confidence.

DATASET OVERVIEW:

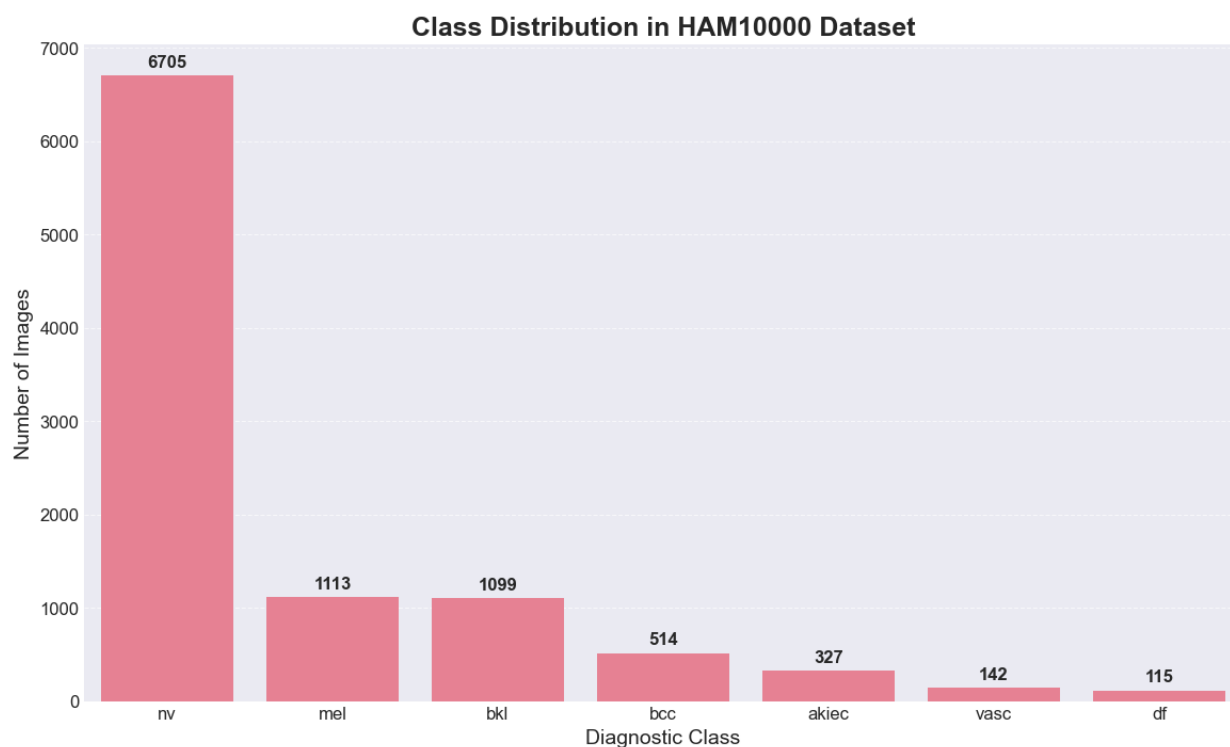
Our project is based on the HAM10000 dataset (Human Against Machine with 10,000 training images), a popular and carefully selected set of dermoscopic photos meant for the classification of common pigmented skin lesions. Clinically significant metadata, including lesion ID, image ID, diagnosis (dx), diagnosis type (dx_type), patient age, sex, and lesion location, are included with each of the 10,015 high-quality dermoscopy images.

We found 57 missing values during the preliminary analysis, the majority of which were in the age column. During preprocessing, these were handled properly. Although PAD-UFES-20, a secondary dataset, was taken into consideration for inclusion, it was not included in this phase because of a large amount of missing data and to guarantee a controlled and targeted scope for model development.

EDA (Exploratory Data Analysis) :

We have performed A thorough EDA on the HAM10000 metadata to uncover patterns and biases. Key findings include:

Class Imbalance

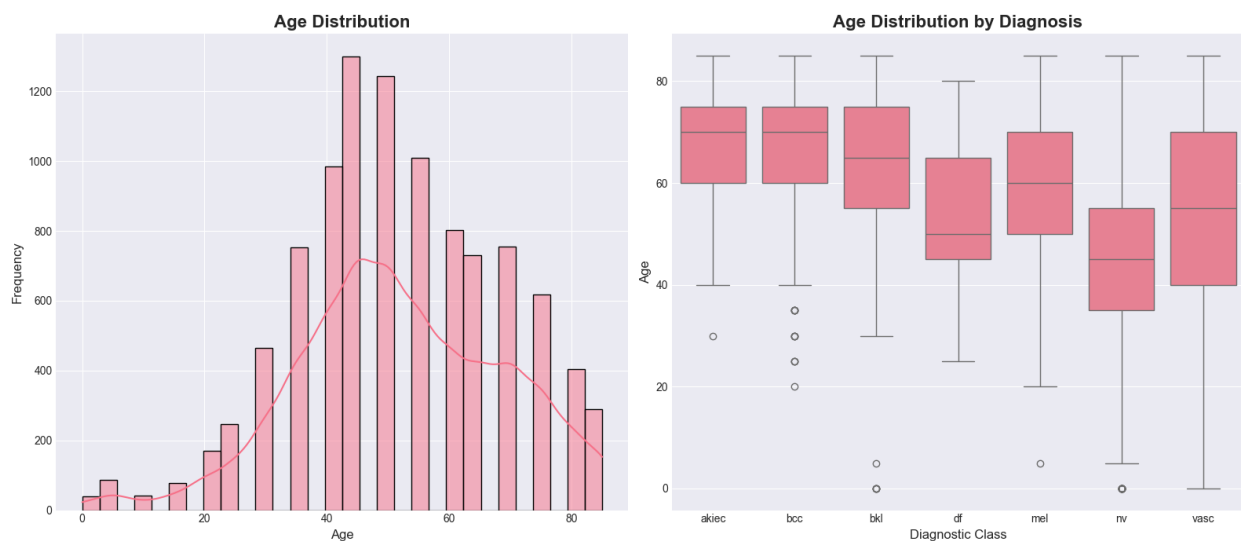


- This bar chart shows a clear class imbalance in the HAM10000 dataset. The majority of the images belong to the 'nv' (nevus) class, which has 6705 images, making up over two-thirds of the dataset. In comparison, all other classes have significantly fewer images.
- The next most common classes are melanoma (mel) and benign keratosis-like lesions (bkl), with just over 1100 images each. From there, the counts drop steeply: bcc (514), akiec (327), vasc (142), and df (115).
- This imbalance means models trained on this data may become biased toward the majority class unless proper techniques like class weighting, data augmentation, or oversampling are applied. Addressing this skew is crucial for building a fair and effective classifier.

VISUALIZATIONS :

Some of the visualizations include the distribution of the key features ,

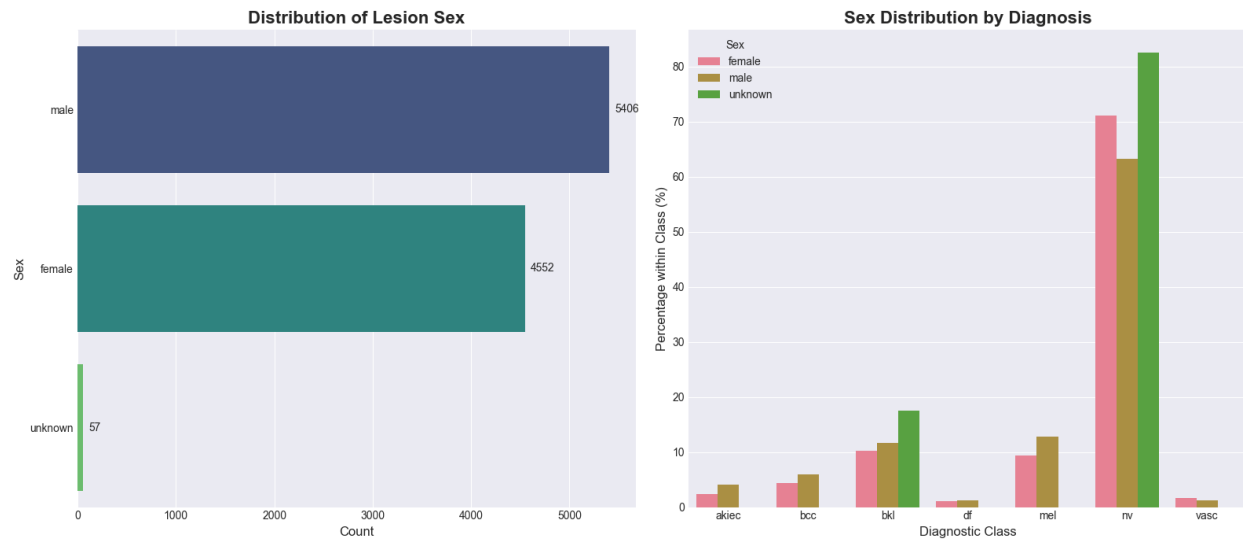
- **Distribution of Age and Age with diagnosis :**



The general age distribution of the patients in the HAM10000 dataset is displayed in the histogram on the left. The majority of cases occur in people between the ages of 40 and 70, with a noticeable peak in the mid-40s. Patients who are very young are uncommon, and after the age of 70, the frequency gradually declines. This indicates that, as is common for skin-related conditions, the dataset's most represented demographic is middle-aged to older adults.

On the right, the boxplot illustrates **age distribution by diagnosis class**. Most diagnostic classes, such as *akiec*, *bcc*, and *mel*, are skewed toward **older individuals**, typically in their 60s and 70s. In contrast, *nv* and *bkl* have a broader age range with more younger patients included. Some outliers appear in all categories, including a few samples from very young patients, even infants. These variations in age by diagnosis could inform age-aware modeling or targeted screening strategies.

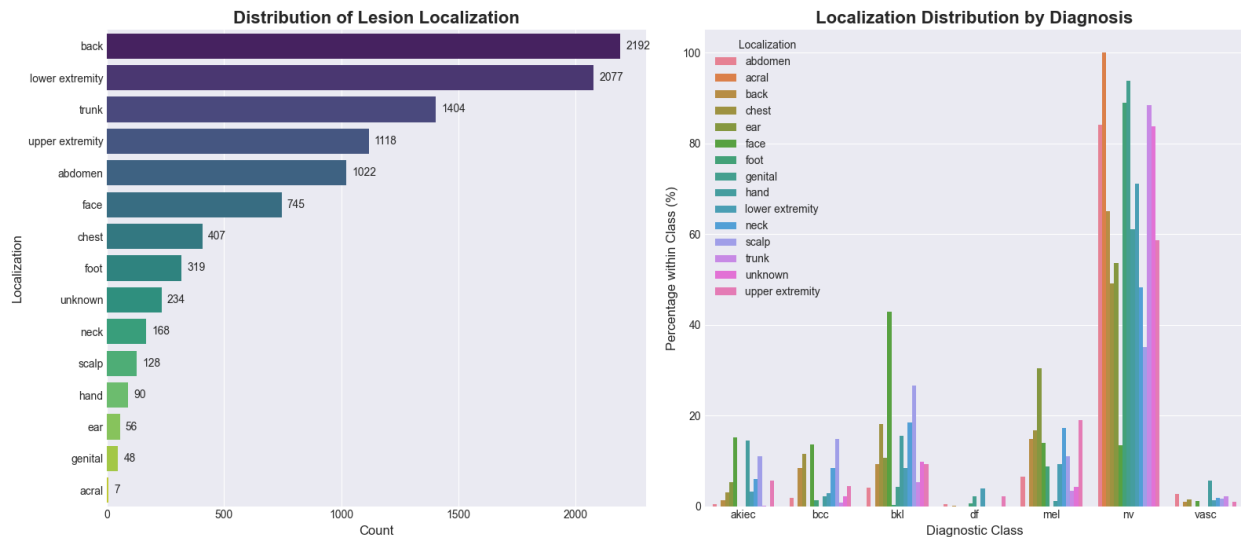
- **Distribution of Lesion sex and its diagnosis:**



The overall gender distribution in the dataset is shown in the chart on the left. As we can see, there are 5,406 more male patients in the data than female patients (4,552). Gender information is missing in 57 cases, which is a small number. Overall, this suggests that the dataset is fairly balanced in terms of gender, despite a small overrepresentation of male patients.

The chart on the right offers a more detailed breakdown, showing how gender is distributed within each specific diagnostic category. It's interesting to note that some conditions appear more frequently in one gender than the other. For instance, a higher proportion of cases for melanoma (mel) and nevi (nv) are from female patients. In contrast, conditions like actinic keratoses (akiec) and basal cell carcinoma (bcc) seem to be slightly more common in males. It's also worth pointing out that the 'Nevus' category contains nearly all the cases with unknown gender, which might indicate that this data was less consistently recorded for this very common lesion type. These kinds of patterns could provide useful clinical insights and may be valuable for building more nuanced, gender-aware predictive models.

- **Distribution of Localization and it's diagnosis class:**

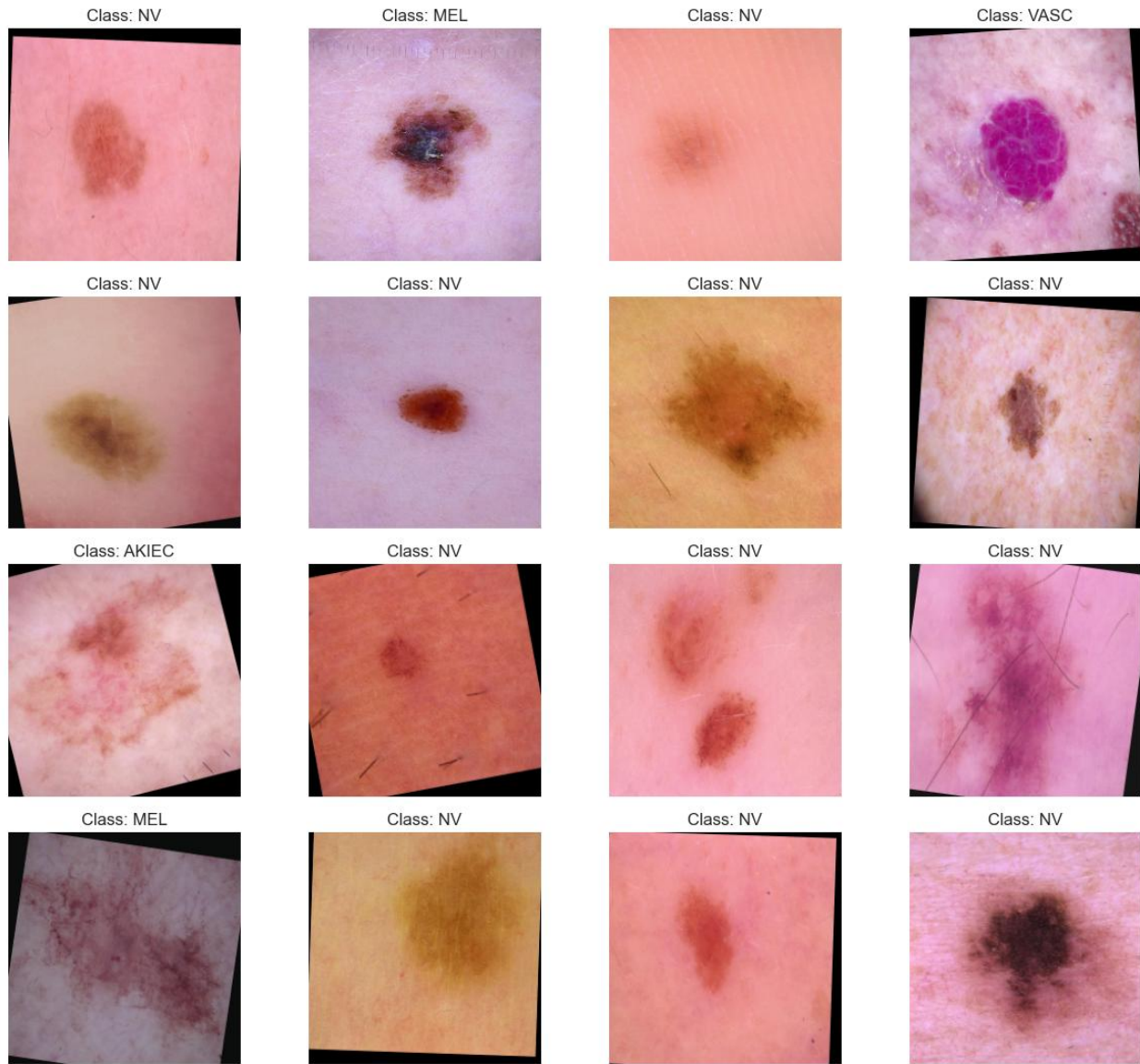


The location of the skin lesions in the dataset is displayed in the bar chart on the left. Certain body parts are obviously much more frequently found than others. With 2,192 photos, the back is the most frequently photographed area. The arms and lower legs come in second with 2,077 photos, and the main body or trunk comes in first with 1,404 photos. On the other hand, very few lesions were found on the hands, ears, scalp, or genital area. This trend may be due to the fact that the more popular spots are either sun-exposed or just more likely to have skin problems.

A more detailed view is provided by the chart on the right, which illustrates the locations of particular lesion types. For instance, nevi (nv) and melanoma (mel) are most frequently found on the trunk, lower limbs, and back. However, benign keratosis-like lesions (bkl) are more frequently observed on the face, upper arms, or legs. Interestingly, the face and back are often affected by basal cell carcinoma (bcc) and actinic keratoses (akiec). This reveals that specific skin conditions prefer to manifest on specific body parts, which can help physicians diagnose patients and aid in the development of more intelligent models that account for location.

- **Visualizing dataset (Batch and Augmented) :**

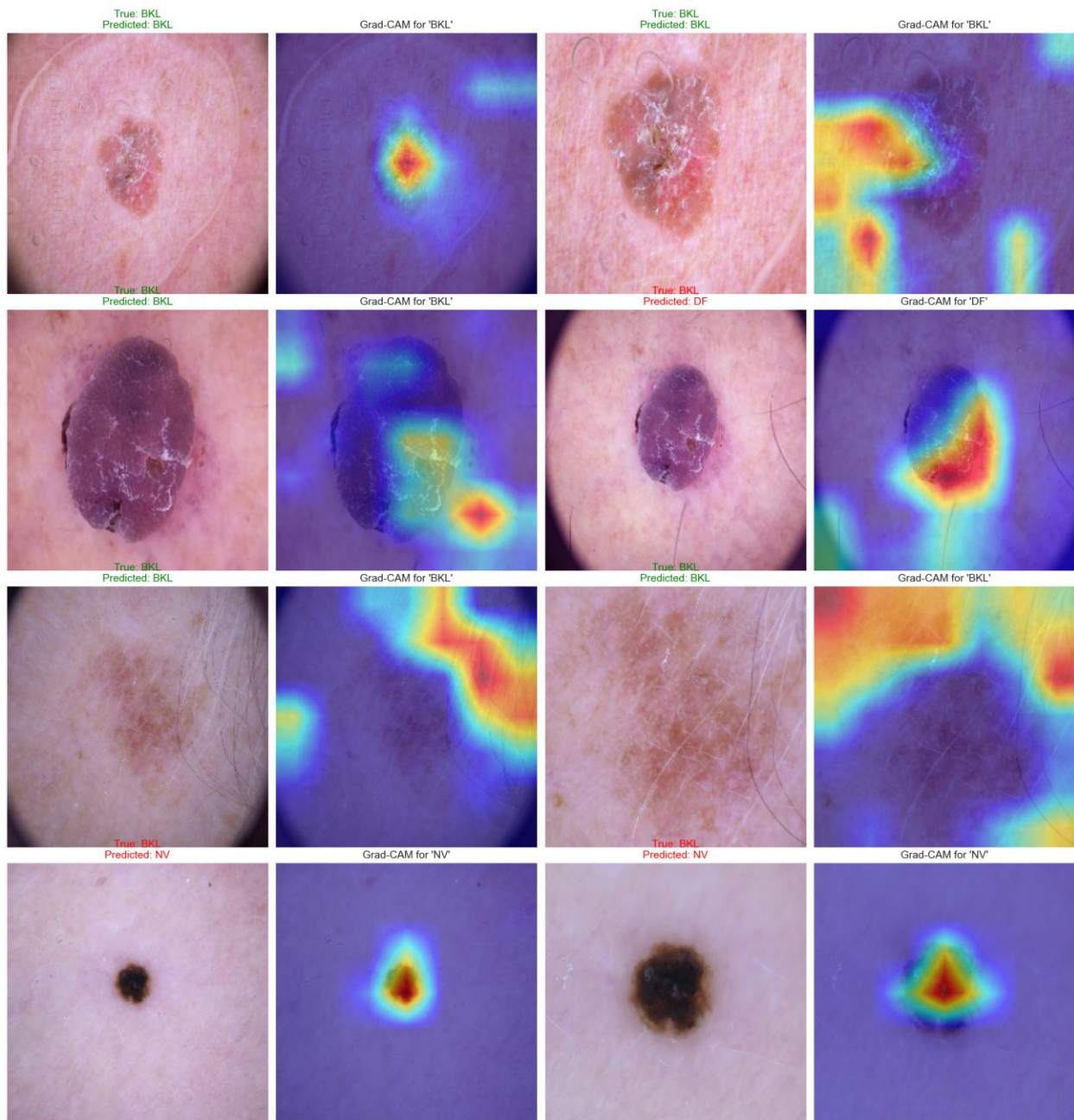
Sample Batch from Training DataLoader (Augmented)



The above image matrix is a sample of Augmented data images from the training data loader indicating an image from each class and the augmented of some class.

Grad-cam Visualization of the dataset :

Model Predictions with Grad-CAM Explanations



The Grad-CAM images above help us understand how the model makes decisions by showing which parts of the image it focused on. In correct predictions (green labels), the heatmaps are mostly centered on the lesion, which means the model looked at the right area. But in wrong predictions (red labels), the attention is spread out or focused on the wrong spots, which helps explain why the model got it wrong. This kind of explanation helps

us check if the model is learning the right things and also shows where it might be confused—especially for similar-looking classes like **BKL**, **DF**, and **NV**.

MODEL ARCHITECTURE (Baseline) :

The Multi-Modal Intelligent Dermoscopy Analysis System (MIDAS_V1) is a specialized model developed for the detailed analysis of skin lesions. Its core strategy involves the integration of two key types of information: the visual data from dermoscopy images and the contextual data from patient records. By uniting the visual evidence with relevant clinical details like patient age and the lesion's location, the system can perform a more thorough analysis than is possible when considering either type of information in isolation.

The process starts with the **Image Encoder**, which handles the visual analysis. This component uses the EfficientNet-B0 model, a highly capable convolutional neural network. Having been pre-trained on a vast collection of images, the encoder excels at identifying and extracting important visual features. It examines the dermoscopy image for key characteristics like texture, color variations, and structural patterns, ultimately producing a detailed numerical summary that captures the lesion's appearance.

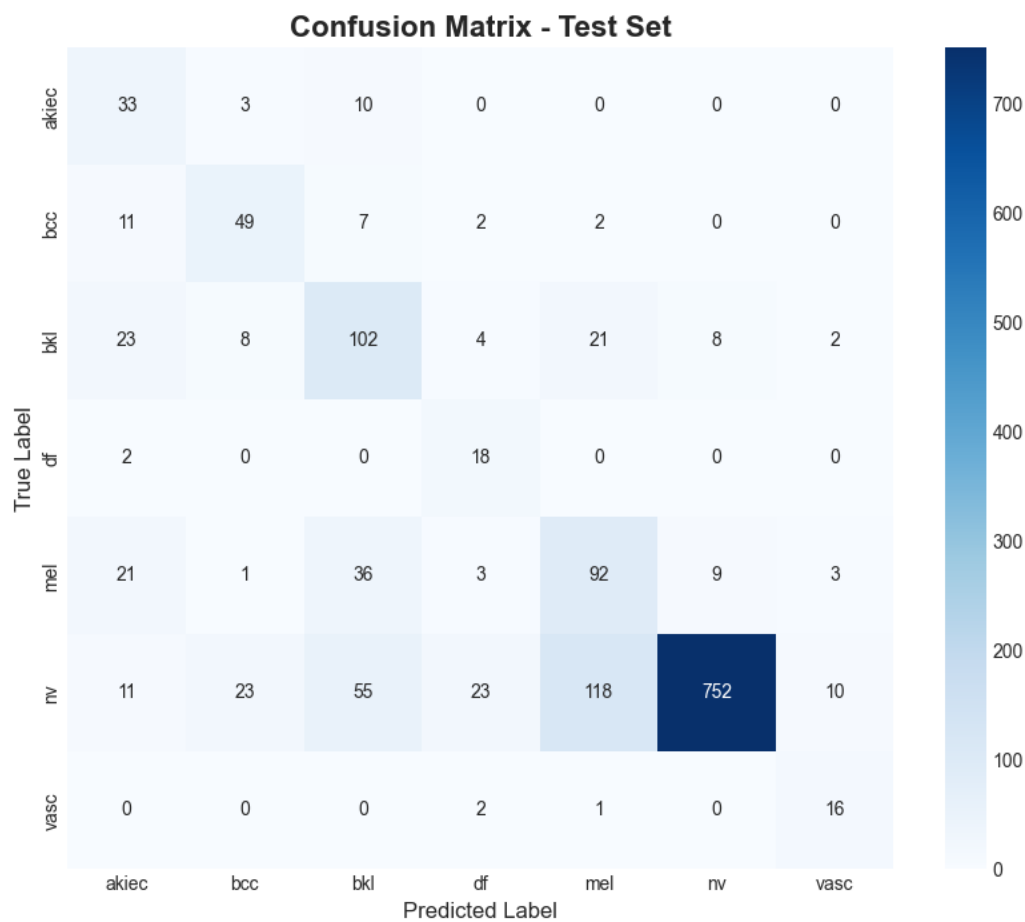
Working in parallel, the **Metadata Encoder** evaluates the patient's accompanying information. This part of the system is a neural network tailored to interpret tabular data, such as the patient's demographic details and the anatomical location of the lesion. Its purpose is to find meaningful patterns within this information and convert the patient's background details into a useful numerical format. In doing so, it supplies the analysis with essential clinical context.

In the final stage, the **Fusion and Classification Head** unites the outputs from the previous two components. The numerical summaries from both the Image Encoder and Metadata Encoder are joined together, creating a single, comprehensive profile of the case. This unified profile is then delivered to the **classifier**, which is the final decision-making part of the network. The classifier evaluates the combined information to arrive at a final determination about the lesion. This integrated method ensures the system's conclusion is based on the full context of the case.

Evaluation metrics (Baseline) :

By Training the model through 50 epochs , we have achieved a 72% test accuracy, which is fairly good.

A confusion matrix to provide a detailed, class-by-class visualization of the model's predictive performance,



from the confusion matrix above, the model does best on the majority class "nv," with 752 correct predictions. However, it has a lot of trouble with minority classes like "df," "akiec," and "vasc," which have high misclassification rates. Significant confusion exists between "mel," "bkl," and "nv," indicating that their visual characteristics overlap. Classes like "bcc" and "bkl" perform moderately, but overall, the pattern shows that the model is biased toward well-represented classes and does not generalize for underrepresented ones. This underscores the impact of class imbalance and the need for specific enhancements in feature discrimination or data augmentation.

INFO: Classification Report on Test Set:

	precision	recall	f1-score	support
akiec	0.33	0.72	0.45	46
bcc	0.58	0.69	0.63	71
bkl	0.49	0.61	0.54	168
df	0.35	0.90	0.50	20
mel	0.39	0.56	0.46	165
nv	0.98	0.76	0.85	992
vasc	0.52	0.84	0.64	19
accuracy			0.72	1481
macro avg	0.52	0.72	0.58	1481
weighted avg	0.80	0.72	0.74	1481

The model achieves a solid **overall accuracy of 72%**, mainly driven by strong performance on the majority class **'nv' (F1-score: 0.85)**. Minority classes like **'akiec'** and **'df'** show high recall but low precision, indicating frequent misclassifications. The **macro F1-score of 0.58** highlights the imbalance in per-class performance. While the model is reliable for dominant classes, it underperforms on less frequent ones.

Improved model architecture:

For our advanced model, we designed a new architecture called MIDAS_V1, which improves upon our baseline by combining image and metadata features more effectively. During experiments, we can easily test different backbones because the model uses a dynamic image encoder (backbone) that was chosen from popular architectures using the timm library. The model learns the relationship between the two inputs by processing metadata independently through a small feedforward network and then fusing it with image features using a cross-attention module.

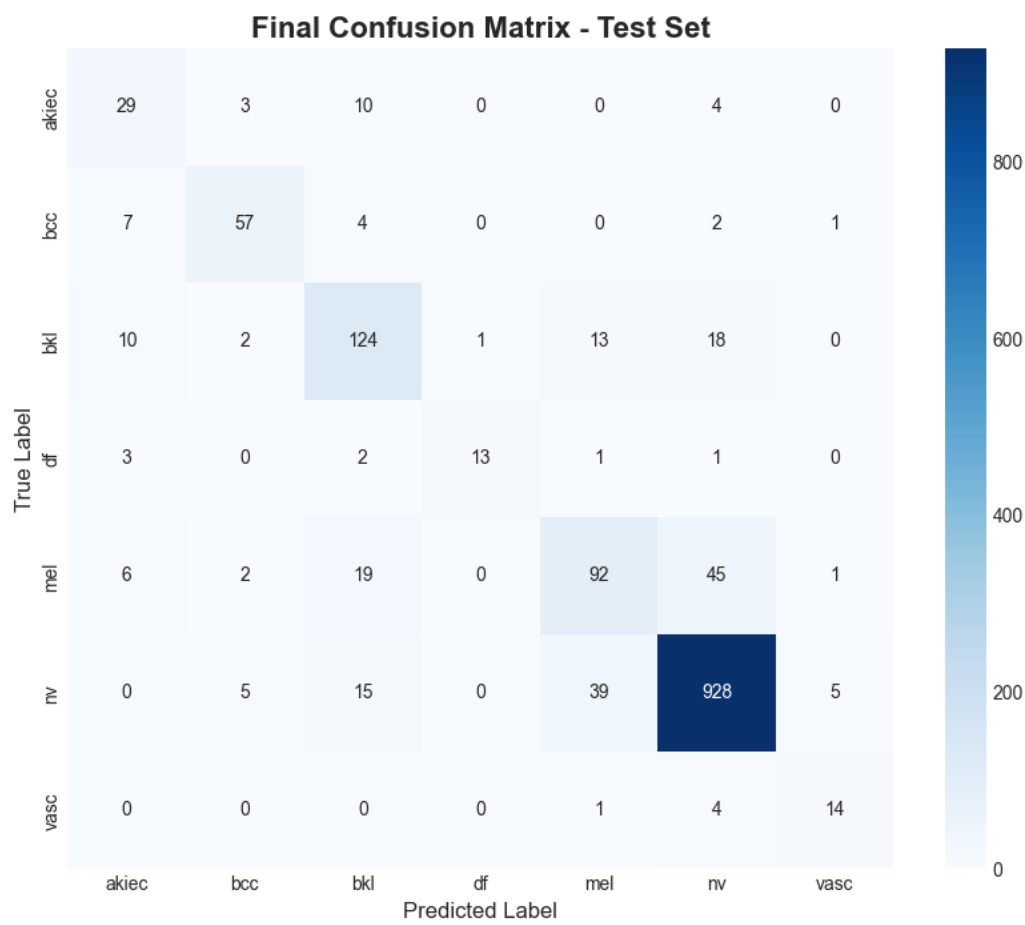
We use the Optuna framework to find the optimal configuration. It assists us in automatically determining the optimal image backbone and optimizing important hyperparameters such as attention dimensions, learning rate, and dropout rate. By modifying its structure and

settings in response to validation results, our model becomes more adaptable and better suited to the dataset.

Evaluation metrics (Improved model) :

From training the improved model for the same 50 epochs, we have achieved a solid test accuracy of around 84%.

A confusion matrix to provide a detailed, class-by-class visualization of the model's predictive performance,



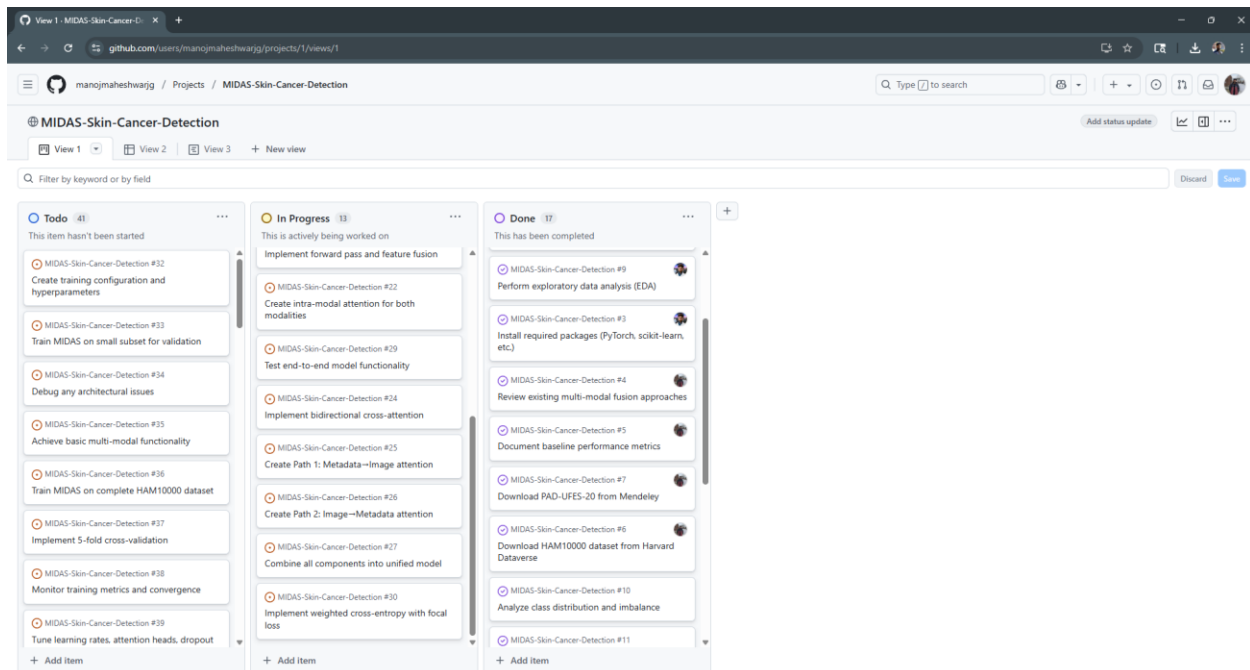
From the confusion matrix above we can infer that the model's performance has clearly improved, as evidenced by the confusion matrix, which shows better predictions for the majority of classes. With 928 accurate predictions, the dominant class "nv" outperforms other classes like "bcc" (57) and "bkl" (124). Better feature learning is indicated by the decrease in misclassifications, particularly between similar classes like mel, bkl, and nv.

Even uncommon classes like "df" and "vasc" experience modest increases. All things considered, the model is more balanced and accurate, making fewer errors overall.

INFO: Final Classification Report on Test Set:				
	precision	recall	f1-score	support
akiec	0.5273	0.6304	0.5743	46
bcc	0.8261	0.8028	0.8143	71
bkl	0.7126	0.7381	0.7251	168
df	0.9286	0.6500	0.7647	20
mel	0.6301	0.5576	0.5916	165
nv	0.9261	0.9355	0.9308	992
vasc	0.6667	0.7368	0.7000	19
accuracy			0.8488	1481
macro avg	0.7454	0.7216	0.7287	1481
weighted avg	0.8485	0.8488	0.8478	1481

The final classification report shows that the model is doing a much better job overall, with an accuracy of **84.88%**. It performs very well on the most common class '**nv**', and also gives good results on other classes like '**bcc**', '**bkl**', and even the rare class '**df**'. Some classes like '**mel**' are still a bit harder for the model, but the scores have improved. The average scores show that the model is now handling all classes more evenly, not just the common ones. This means the model is more reliable and balanced than before.

Git-projects (PM Tool):



Git-Who:

Conclusion:

So far we have completed all the steps up to building the improved MIDAS V1 model. This new version combines image and metadata features using attention, and it gives better results than our first (baseline) model.